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1. Let's consider the points $(a_i \mod 2, b_i \mod 2, c_i \mod 2)$. We know that each of those coordinates are 0 or 1, so there are $2^3 = 8$ possible combinations for the coordinates for each i, but we have nine points. By the Pigeonhole Principle, there must be at least two points (a_j, b_j, c_j) and (a_k, b_k, c_k) such that $(a_j \mod 2, b_j \mod 2, c_j \mod 2) = (a_k \mod 2, b_k \mod 2, c_k \mod 2)$. This means $a_j \equiv a_k \pmod 2$, $b_j \equiv b_k \pmod 2$, and $c_j \equiv c_k \pmod 2$. Therefore, $\frac{a_j+a_k}{2}$, $\frac{b_j+b_k}{2}$, and $\frac{c_j+c_k}{2}$ are integers, and $(\frac{a_j+a_k}{2}, \frac{b_j+b_k}{2}, \frac{c_j+c_k}{2})$ is a point with integer coordinates. □

2. For this problem, we can use the pigeonhole principle. Let's say every pair of numbers $(1, 2), (3, 4), \ldots, (99, 100)$ is a hole, and the pigeons are the 51 positive integers. Since there are 50 of those pairs, there must be at least two integers in one pair, which means those two are consecutive. □

3. Let's start with knights. The optimal way to place knights so that none of them attack each other is on all the squares of the same color, since a knight will never attack another one that is on the same colored square. That means the answer is 32.

To prove that we cannot have more than 32 knights on the chessboard without them attacking each other, let's split the board into eight 4×2 rectangles. Since we are distributing more than 32 knights among eight rectangles, by the Pigeonhole Principle, at least one rectangle must have more than four knights. But this cannot be possible, because then at least one pair of knights in that rectangle would be attacking each other. Therefore, 32 is the maximum.□

For bishops, an optimal way to place them is to keep eight on the bottommost row and six more centered at the top row. This gives us 14 bishops.

Now, let's prove that we cannot have more than 14 bishops on the chessboard without them attacking each other. We know that there are 14 diagonals on the chessboard totally, so if there are more than 14 bishops, by the Pigeonhole Principle, there must be at least two bishops on the same diagonal. But this can't be possible, because then they would be attacking each other. Therefore, 14 is the greatest number of bishops we can have. □

4. Since S has 10 elements, it has $2^{10} - 1 = 1023$ nonempty subsets. Now, let $s(A)$ denote the sum of the integers in set A. That means $1 \leq s(T) \leq 91 + 92 + 93 + \ldots + 100 = 955$, and $1 \leq s(U) \leq 955$. Since there are 1023 subsets and (at most) 955 possible sums, at least two distinct subsets T and U must have the same sum. (If T and U are not disjoint, we can subtract whatever common terms there are, and the result would be two disjoint subsets of S that have the same sum.) □

5. Let's consider the $n + 1$ numbers $1, 11, 111, \ldots, 1\ldots1$(with n + 1 digits). When we divide each of those by n, we have n possible remainders. Since we have $n + 1$ numbers, by the Pigeonhole Principle, at least two of those numbers must have the same remainder when divided by n. Let's say those two numbers have a digits of 1 and b digits of 1 respectively. If we subtract them, since they have the same remainder when divided by n, the difference is a multiple of n which only consists of 1's and 0's. □

6. Each person can have at most $n - 1$ friends, so the number of friends someone can have ranges from 0 to $n - 1$. However, if someone has $n - 1$ friends, then another person cannot have 0 friends, because they have to be friends with the first person. By similar logic, if someone has any less than $n - 1$ friends, another person cannot have $n - 1$ friends. In any case, that leaves us with just $n - 1$ options, and n people. So by the pigeonhole principle, two people must have the same number of friends. □

7. We know that if we are given three points, we can create a circle that passes through all of them. Since all three points are on the surface of the sphere, the circle is on the sphere as well. So you can create two hemispheres of the sphere using that circle. If both of the other points are on one side, you can choose the side of the hemisphere that includes those points. If they are both on different sides of the circle, you can point the hemisphere in one of those directions and at least 4 of the total points will be on that hemisphere. □

8. We can split this into two cases:

Case 1: at least two of a_i are the same. Let's call those two numbers a_j and a_k. Since one of the terms in the product is $(a_j - a_k)$, the total product must be 0, which is a multiple of $n!$.

Case 2: all a_i are distinct. For this case, let's consider the $n+1$ numbers $a_0, a_1, \ldots a_n$ modulo p, where p is a prime divisor of $n!$. Since there are at most n possible values of p, by the Pigeonhole Principle, there must be at least two indices i and j such that $a_i \equiv a_j \pmod{p}$. This means $a_i - a_j \equiv 0 \pmod{p}$. Since we can apply this to every prime divisor of $n!$, we know that $n!$ divides the whole product. $\square$

9a. Let's consider the number $(k+1)!$. When we add any integer from 2 to $k+1$ $(k+1)!$, we get a composite number, because we can factor out whichever integer we added from both numbers. That means $(k+1)!+2$, $(k+1)!+3$, ..., and $(k+1)!+(k+1)$ are all composite. Since the list of integers from 2 through $k+1$ consists of k numbers, we have found k consecutive positive integers that are all composite. $\square$

9b. For a number to be rational, it must terminate or repeat when written as a decimal. Let's consider the partial sums S_n where $S_n = \sum_{p \leq n \text{ prime}} \frac{1}{2^p}$. If we represent that in binary, each term $\frac{1}{2^p}$ contributes a 1 at the p^{th} place after the binary point. That means the infinite sum would look like $0.01101010001\ldots$ written in binary.

We know that there are infinite primes, so this number cannot end at any point. That means, for it to be rational, it must terminate.

But it cannot repeat either. This is because, as we proved in the last part, we can always find k consecutive composite integers for any integer k. So if we assume it starts repeating at some point such that the longest string of consecutive composite integers is k, we could always just wait until we find another string of $k+1$ consecutive composite integers. This means it cannot repeat at any point.

Since this number cannot terminate or repeat at any point, it is irrational. $\square$